

ROHAN CHICKALKAR

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Professional Summary:

- **Data & GenAI** Engineer with 7+ years designing large-scale **ETL**, **data lake**, and **RAG** systems across **AWS**, **Azure**, **GCP**, and **Databricks**, delivering pipelines processing 50M+ records across retail, finance, and healthcare.
- Built 20+ enterprise **ETL** workflows using **Spark**, **Glue**, **Dataflow**, **Data Factory**, and **Delta Lake**, enabling reliable ingestion of 10M–40M monthly operational and transactional events.
- Engineered 150+ ML features on **Databricks** using **Spark**, **Delta Lake**, and **MLflow**, supporting 5 enterprise credit-risk and fraud-detection models with full lineage and experiment tracking.
- Delivered 3 production **RAG** platforms using **Vertex AI**, **Embeddings**, **Vector Search**, and **BigQuery**, indexing 30K documents and supporting 1M+ annual mechanic and operations queries.
- Designed 12+ domain data models on **BigQuery**, **Redshift**, and **Snowflake**, optimizing analytics for more than 2TB monthly data growth and improving query speeds by 30%.
- Developed 6 high-throughput microservices using **Cloud Run**, **Lambda**, and **Azure Functions**, routing 500K+ daily events with sub-900 ms latency to support diagnostic and retail analytics workloads.
- Implemented 25+ data-quality rule sets detecting 30K+ anomalies monthly across regulated domains, improving downstream reliability for pipelines spanning 4 global regions.
- Automated 500K daily incremental updates using **S3**, **Cloud Storage**, and **Delta Lake**, ensuring SLA-bound delivery with 99.9%+ success across 8 business ingestion domains.
- Built 10 ML monitoring dashboards using **BigQuery**, **MLflow**, and **Looker**, tracking drift and quality metrics across 3K daily inference events for strict governance and audit needs.
- Collaborated with 25+ stakeholders across engineering, analytics, and ML teams to align **RAG**, **ETL**, and **feature store** logic with 15 compliance-bound business requirements.

Technical Skills:

- **Programming Languages:** Python, Java, Scala, SQL, R, Shell Scripting
- **Big Data Technologies:** Apache Spark, Hadoop, Hive, Sqoop, Pig, HBase, Kafka
- **GenAI & LLM Technologies:** LLMs, GPT, LLaMA, Prompt Engineering, RAG, Embedding Models, Vector Search, Vector DBs, LangChain, LangGraph, OpenAI API, HuggingFace, Transformers, Vertex AI GenAI Studio
- **Cloud Platforms:**
 - AWS (S3, Glue, Lambda, Redshift, EMR)
 - Azure (Data Factory, Functions, Monitor)
 - GCP (BigQuery, Dataflow, Composer, Vertex AI, Cloud Run)
- **Data Modeling & Databases:** OLTP/OLAP Modeling, Star/Snowflake Schema, Normalization, BigQuery, Snowflake, Redshift, Azure SQL, MySQL, PostgreSQL
- **ETL Tools:** Glue, Dataflow, Data Factory, Databricks Workflows/Jobs, Cloud Composer, Airflow
- **Database Architectures:** Data Lake, Lakehouse, Delta Lake, Data Warehouse, MPP Systems, Columnar Storage, Partitioning & Clustering
- **Data Processing Libraries:** PySpark, Pandas, NumPy, Delta Lake APIs, Beam (Dataflow), Spark SQL
- **Machine Learning & Graph Processing:** Scikit-learn, XGBoost, MLflow, Feature Stores, Vertex AI Training/Endpoints, GraphFrames, NetworkX
- **BI & Reporting Tools:** Power BI, Tableau, AWS QuickSight, Looker, BigQuery BI Engine

- **APIs & Authentication:** REST APIs, OAuth2, JWT, API Gateway, Cloud Endpoints, Secrets Manager, Key Management (KMS)
- **Development Methodologies:** Agile, Scrum, Kanban, Test-Driven Development (TDD), MLOps Best Practice
- **CI/CD & Version Control:** Git, GitHub, GitLab, Jenkins, Azure DevOps, Cloud Build, Docker

Professional Experience:

GenAI Engineer | Toyota Motor Corp, USA

Aug 2024 – Present

Project: GenAI Chat Assistant for Vehicle Diagnostics & Troubleshooting (Vertex AI)

- Designed a **Vertex AI RAG** pipeline processing **5M** repair and diagnostic logs, enabling mechanics to query faults, symptoms, and resolutions using natural language.
- Generated **30K** document embeddings using **Vertex AI Embeddings** and stored them in **Vector Search**, improving retrieval accuracy for **12K** service terms.
- Ingested **2TB** historical sensor and telematics data into **BigQuery**, powering LLM-backed troubleshooting recommendations.
- Built **6 Cloud Run** microservices for query routing, context assembly, and response ranking, maintaining sub-**900** ms latency.
- Created **18** prompt templates optimizing LLM responses for diagnostics, repair steps, error codes, and part replacements across **4** vehicle platforms.
- Constructed **10** data-quality rules detecting **20K** malformed logs monthly before embedding ingestion.
- Implemented **3** multi-turn dialogue flows with **Vertex AI Endpoints**, improving continuity across **8** mechanic interaction scenarios.
- Automated daily refresh of **500K** new service records via **Cloud Composer**, ensuring latest troubleshooting content remains indexed.
- Tuned vector similarity search for **1M** queries/month, improving retrieval precision by **25%** and reducing hallucination risk.
- Built **4** guardrail layers enforcing PHI-safe and safety-critical response boundaries with **<1%** rejection error rate.
- Logged **3K** model inference events daily into **BigQuery** for drift monitoring and regulatory auditability.
- Collaborated with **9** service engineers to align LLM outputs with OEM diagnostic standards and **15** compliance constraints.

Data/ML Engineer | Centene, USA

Dec 2023 - June 2024

Project: Credit Risk Feature Engineering & Scoring Pipeline on Databricks using the required framework.

- Engineered **150** credit-risk features using **Databricks**, **Spark**, and **Delta Lake**, covering utilization, delinquency, and payment trends for **5** enterprise scoring models.
- Processed **40M** daily account and transaction rows through **Spark ETL** pipelines with consistent sub-**25** minute runtimes across **3** feature layers.
- Built **18** reusable feature notebooks orchestrated through **Databricks Jobs**, supporting daily and intraday model refresh cycles.
- Implemented **25** data-quality checks catching **30K** monthly anomalies in balances, credit limits, and payment histories before landing in **Delta Lake**.
- Created **6** incremental merge workflows using **Delta Lake MERGE INTO** logic, updating **12M** account records with **<5** minute latency.

- Logged **3K** model-training artifacts using **MLflow**, improving reproducibility and traceability for risk model experiments.
- Designed **10** time-window aggregations capturing **90, 180, and 360** day risk patterns for enhanced feature stability.
- Integrated **4** upstream financial systems delivering **8M** daily events into curated **Delta Lake** gold tables.
- Tuned **Spark** partitions to handle **2TB** monthly growth, producing **20%** faster feature-generation cycles.
- Automated **5** validation reports summarizing feature drift, missingness, and distribution shifts across **3** environments.
- Optimized **Spark** joins for **50M** row datasets, reducing compute cost by **30%** using broadcast strategies.
- Collaborated with **7** data scientists to align feature logic with regulatory and model-governance criteria across **15** attributes.

Data Engineer | Amazon, India

May 2020 – July 2022

Project: Retail Order Lifecycle Lakehouse on AWS using the required framework.

- Designed an **AWS ETL** pipeline ingesting **30M** monthly order, shipment, and return events into **S3, Glue, and Redshift**, enabling unified visibility for **8** retail domains.
- Engineered **15 Glue** jobs to standardize order status, cancellation codes, and delivery promises, delivering consistent sub-**20** minute batch loads.
- Processed **500K** daily incremental order updates using **Lambda** and **S3** event triggers, ensuring accurate near-real-time lifecycle tracking.
- Modeled **12** fact and **20** dimension tables in **Redshift** to support analytics teams querying **60M** rows/day across fulfillment and CX workflows.
- Implemented reconciliation logic detecting **25K** discrepancies monthly across OMS, transportation, and carrier systems.
- Built **6** SLA-bound pipelines in **Glue** maintaining **99.9%** on-time delivery of curated datasets to downstream users.
- Automated **3** retry and fallback mechanisms handling **10K** transient ingestion failures weekly without manual intervention.
- Tuned **Redshift** sort and dist keys for **2TB** monthly data growth, improving query performance by **35%**.
- Consolidated **40K** daily shipments from **5** carrier feeds using standardized schemas and automated code mappings.
- Integrated **7** retail systems, including OMS, WMS, and CX tools, enabling consistent order lifecycle alignment across **4** regions.
- Added **25** data-quality rules catching **18K** anomalies per cycle across address, item, and promise-level attributes.
- Partnered with **9** cross-functional teams to refine order-lifecycle logic and streamline delivery-exception reporting across **3** fulfillment networks.

Data Engineer | Cognizant, India

Dec 2018 – Oct 2020

Project: Healthcare Claims Ingestion & Standardization on GCP

- Built a **GCP ETL** pipeline ingesting **20M** monthly claims through **Cloud Storage, Dataflow, and BigQuery**, enabling curated outputs for **6** healthcare domains.
- Engineered **12** ingestion templates in **Dataflow** to standardize institutional, professional, and pharmacy claims with consistent sub-**15** minute loads.
- Processed **500K** daily JSON, CSV, and X12 files via **Cloud Storage** triggers, ensuring validated, PHI-safe downstream delivery.
- Designed **25** business-rule transformations in **Dataflow** to normalize diagnosis, procedure, and billing codes across **3** payer systems.
- Built **10** dimensional tables and **8** fact layers in **BigQuery**, supporting analytics teams querying **50M** rows/day.
- Implemented automated anomaly checks detecting **30K** data-quality exceptions monthly across eligibility, provider, and claims datasets.

- Integrated **4** external payer sources using standardized schemas, reducing onboarding effort by **40%**.
- Automated **5** domain-specific reconciliation checks, resolving **8K** mismatches per cycle.
- Tuned **BigQuery** partitions to handle **2TB** monthly growth while achieving consistent **25%** query performance gains.
- Established lineage for **600** assets using **Data Catalog**, improving audit visibility for HIPAA-governed workflows.
- Built **3** SLA-bound pipelines in **Composer** ensuring **99.9%** on-time daily delivery.
- Partnered with **7** cross-functional teams to align claim-standardization logic with **15** regulatory attributes.

Python Developer | Medline, India

Mar 2017 – Nov 2018

Project: Claims Data Processing Pipeline on Azure

- Built an **Azure ETL** pipeline ingesting **10M** monthly claims rows using **Data Factory** and **Python**, improving data readiness for analytic teams across **3** operational units.
- Developed **5** reusable **Python** modules for schema checks, null scans, and type enforcement, reducing manual QA cycles by **30%**.
- Orchestrated **12** ingestion workflows in **Azure Data Factory**, enabling hourly refresh windows with consistent sub-**20** minute runtimes.
- Implemented **Azure Functions** to auto-cleanse **50K** daily malformed records, eliminating **2** recurring production defects.
- Optimized claim parsing logic to handle **100K** complex nested JSON files, reducing pipeline latency by **40%**.
- Created validation rules for **25** business attributes, catching **15K** data-quality exceptions per month before landing.
- Automated audit logging across **4** stages using **Azure Monitor**, enabling traceability for **100%** of pipeline executions.
- Developed a durable retry mechanism processing **5K** transient failures weekly with zero manual intervention.
- Containerized **Python** jobs into **3** deployable units to streamline promotions across **2** environments.
- Tuned **Azure SQL** staging for **1M** upserts/day via partitioning and index optimization, improving write throughput by **35%**.
- Integrated **3** downstream systems through curated datasets refreshed every **4** hours.
- Coordinated with **6** cross-functional stakeholders to align validation logic with **5** compliance requirements.

Certifications:

- Google Cloud Certified Professional Data Engineer
- Databricks Certified: Data Engineer Associate
- Microsoft Certified Fabric Data Engineer Associate
- AWS Generative AI Applications Professional Certificate
- IBM Data Science Professional Certificate
- Microsoft Certified Azure AI Fundamentals

Education:

Texas A&M University | Commerce, TX

Master's in business Analytics & Artificial Intelligence, Major in Data Science

GPA 3.70, Dean's Excellence Scholarship

Publications:

Employee Attrition Using Machine Learning Algorithms — Springer, Proceedings of International Conference on Data Science and Applications, Lecture Notes in Networks and Systems, vol.288, Singapore, 2022